

Homework 11

Solution

1. Let $f(x) = e^{\sin x}$ for every $x \in \mathbb{R}$.

Compute the first three Taylor polynomials of f about $x_0 = 0$, that is, $T_0(x)$, $T_1(x)$ and $T_2(x)$.

Solution:

We note that for every $x \in \mathbb{R}$ we have

$$f^{(0)}(x) = e^{\sin x}$$

$$f^{(1)}(x) = e^{\sin x} \cdot \cos x$$

$$f^{(2)}(x) = e^{\sin x} \cos x \cdot \cos x + e^{\sin x} \cdot (-\sin x) = e^{\sin x} (\cos^2 x - \sin x)$$

Thus for every $x \in \mathbb{R}$ we have

$$T_2(x) = \sum_{n=0}^2 \frac{f^{(n)}(0)}{n!} x^n = \frac{e^{\sin 0}}{0!} x^0 + \frac{e^{\sin 0} \cos(0)}{1!} x^1 + \frac{e^{\sin 0} (\cos^2 0 - \sin 0)}{2!} x^2 = 1 + x + \frac{x^2}{2}$$

and therefore

$$T_1(x) = 1 + x, \quad T_0(x) = 1$$

2. (a) Consider $q = 1 - \frac{1}{2!} + \frac{1}{4!} - \frac{1}{6!} = \frac{389}{720}$.

Prove that

$$|\cos(1) - q| \leq \frac{1}{8!}$$

Solution:

We know that given a Leibniz series, it holds that

$$\left| \sum_{n=k}^{\infty} (-1)^{n+1} a_n \right| \leq a_k$$

We know that

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad \forall x \in \mathbb{R}$$

thus

$$\cos 1 = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} = 1 - \frac{1}{2!} + \frac{1}{4!} - \frac{1}{6!} + \frac{1}{8!} - \frac{1}{10!} + \dots$$

This is clearly a Leibniz series, thus

$$|\cos(1) - q| = \left| \frac{1}{8!} - \frac{1}{10!} + \dots \right| \leq \frac{1}{8!}$$

- (b) Compute the value of $\sqrt[5]{33}$ by using Taylor approximation to an error smaller than $\frac{3}{1024000}$.

Solution:

Consider the function defined by $f(x) = x^{\frac{1}{5}}$ for every $x > 0$ and choose $x_0 = 32$.

We now compute:

$$\begin{aligned}f'(x) &= \frac{1}{5}x^{-\frac{4}{5}} \Rightarrow f'(32) = \frac{1}{80} \\f''(x) &= -\frac{4}{25} \cdot x^{-\frac{9}{5}} \Rightarrow f''(32) = -\frac{1}{3200} \\f'''(x) &= \frac{36}{125}x^{-\frac{14}{5}}\end{aligned}$$

By the Taylor Remainder theorem, there exists a point $32 < c < 33$ for which

$$|R_2(33)| = \frac{|f'''(c)|}{3!} \cdot (33 - 32)^3 = \frac{36}{125 \cdot 6} c^{-\frac{14}{5}} = \frac{36}{125 \cdot 6 \cdot c^{\frac{14}{5}}} \stackrel{c > 32}{<} \frac{36}{125 \cdot 6 \cdot 16384} = \frac{3}{1024000}$$

We note that

$$T_2(x) = f(32) + f'(32)(x - 32) + \frac{f''(32)}{2}(x - 32)^2$$

Plugging the values of the derivatives into the formula, we get

$$T_2(33) = 2 + \frac{1}{80} - \frac{1}{2 \cdot 3200}$$

We conclude that

$$\left| \sqrt[5]{33} - \left(2 + \frac{1}{80} - \frac{1}{2 \cdot 3200} \right) \right| < \frac{3}{1024000}$$

(c) Let $I = \int_0^1 e^{-x^2} dx$. Find a rational number $q \in \mathbb{Q}$ for which

$$|I - q| \leq \frac{1}{9 \cdot 4!}$$

Solution:

First we represent I as a series using the Taylor expansion of e^x

$$I = \int_0^1 e^{-x^2} dx = \int_0^1 \left(\sum_{n=0}^{\infty} \frac{(-x^2)^n}{n!} \right) dx = \sum_{n=0}^{\infty} \int_0^1 (-1)^n \cdot \frac{x^{2n}}{n!} dx = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \cdot \left[\frac{x^{2n+1}}{2n+1} \right]_0^1 = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)n!}$$

Thus

$$I = 1 - \frac{1}{3 \cdot 1!} + \frac{1}{5 \cdot 2!} - \frac{1}{7 \cdot 3!} + \frac{1}{9 \cdot 4!} - \dots$$

This is clearly a Leibniz series, thus

$$\left| \frac{1}{9 \cdot 4!} - \dots \right| \leq \frac{1}{9 \cdot 4!}$$

We write

$$q = 1 - \frac{1}{3 \cdot 1!} + \frac{1}{5 \cdot 2!} - \frac{1}{7 \cdot 3!}$$

we note that $q \in \mathbb{Q}$ by the closure of $\mathbb{Q}$. Thus

$$|I - q| \leq \frac{1}{9 \cdot 4!}$$

3. In this question, we will prove the formula for the variance of the Geometric random variable that will be revisited in the course of Introduction to Probability.

Let $0 < p < 1$.

- (a) Prove that $\sum_{n=1}^{\infty} [n(1-p)^{n-1} \cdot p] = \frac{1}{p}$.

Solution:

We know that the series $\sum_{n=1}^{\infty} (1-p)^n$ is a tail of a geometric series, and as $0 < p < 1$, then $0 < 1-p < 1$ which implies that the series is convergent, and

$$\sum_{n=1}^{\infty} (1-p)^n = \sum_{n=0}^{\infty} (1-p)^n - 1 = \frac{1}{1-(1-p)} - 1 = \frac{1}{p} - 1$$

Thus, we can use term-by-term differentiation:

$$-\frac{1}{p^2} = \left[\frac{1}{p} - 1 \right]' = \left[\sum_{n=1}^{\infty} (1-p)^n \right]' \stackrel{\text{T.B.T.D}}{=} \sum_{n=1}^{\infty} [(1-p)^n]' = \sum_{n=1}^{\infty} -n(1-p)^{n-1} = -\sum_{n=1}^{\infty} n(1-p)^{n-1}$$

Thus

$$\sum_{n=1}^{\infty} [n(1-p)^{n-1} \cdot p] = p \sum_{n=1}^{\infty} n(1-p)^{n-1} = p \cdot \frac{1}{p^2} = \frac{1}{p}$$

- (b) Prove that $\sum_{n=1}^{\infty} [n^2(1-p)^{n-1} \cdot p] = \frac{2-p}{p^2}$.

Solution:

By item (a), we have $\sum_{n=1}^{\infty} [n(1-p)^{n-1} \cdot p] = \frac{1}{p}$ and therefore $\sum_{n=1}^{\infty} n(1-p)^{n-1} = \frac{1}{p^2}$.

Differentiating this series again by using term-by-term differentiation, we get

$$\begin{aligned} -\frac{2}{p^3} &= \left[\sum_{n=1}^{\infty} n(1-p)^{n-1} \right]' = \sum_{n=1}^{\infty} [n(1-p)^{n-1}]' = \sum_{n=1}^{\infty} [-n(n-1)(1-p)^{n-2}] = -\sum_{n=1}^{\infty} n(n-1)(1-p)^{n-2} \\ &= -\sum_{n=2}^{\infty} n(n-1)(1-p)^{n-2} = -\sum_{n=1}^{\infty} n(n+1)(1-p)^{n-1} = -\sum_{n=1}^{\infty} n^2(1-p)^{n-1} - \sum_{n=1}^{\infty} n(1-p)^{n-1} \\ &= -\sum_{n=1}^{\infty} n^2(1-p)^{n-1} - \frac{1}{p^2} \end{aligned}$$

Thus

$$\sum_{n=1}^{\infty} [n^2(1-p)^{n-1} p] = p \sum_{n=1}^{\infty} n^2(1-p)^{n-1} = p \cdot \left[\frac{2}{p^3} - \frac{1}{p^2} \right] = \frac{2}{p^2} - \frac{1}{p} = \frac{2-p}{p^2}$$

(c) Conclude that

$$\sum_{n=1}^{\infty} \left[n^2 (1-p)^{n-1} \cdot p \right] - \left[\sum_{n=1}^{\infty} \left[n (1-p)^{n-1} \cdot p \right] \right]^2 = \frac{1-p}{p^2}$$

Solution:

By items (a) and (b) we get that

$$\sum_{n=1}^{\infty} \left[n^2 (1-p)^{n-1} \cdot p \right] - \left[\sum_{n=1}^{\infty} \left[n (1-p)^{n-1} \cdot p \right] \right]^2 = \frac{1-p}{p^2} = \frac{2-p}{p^2} - \left[\frac{1}{p} \right]^2 = \frac{2-p}{p^2} - \frac{1}{p^2} = \frac{1-p}{p^2}$$

4. Consider the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2}$.

(a) Prove that there exists a $K \in \mathbb{R}$ for which

$$K = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2}$$

and that $\frac{8}{9} \leq K \leq 1$.

Solution:

We note that the sequence $\left(\frac{1}{(2n+1)^2}\right)_{n=0}^{\infty}$ is decreasing (1 over increasing and positive) and

$\lim_{n \rightarrow \infty} \frac{1}{(2n+1)^2} = 0$, thus the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2}$ is a Leibniz series, which implies that it converges.

Thus, there exists a $K \in \mathbb{R}$ such that $K = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2}$, and we have $\frac{8}{9} = 1 - \frac{1}{9} \leq K \leq 1$.

(b) Compute the limit $\lim_{x \rightarrow 0^+} [\arctan(x) \cdot \ln(x)]$.

Solution:

$$\lim_{x \rightarrow 0^+} [\arctan(x) \cdot \ln(x)] = \lim_{x \rightarrow 0^+} \left[\frac{\arctan(x)}{x} \cdot x \ln(x) \right] \stackrel{\text{AOL}}{=} 1 \cdot 0 = 0$$

Where

$$\lim_{x \rightarrow 0^+} \frac{\arctan(x)}{x} \stackrel{\frac{0}{0}}{=} \lim_{x \rightarrow 0^+} \frac{\left(\frac{1}{1+x^2}\right)}{1} \stackrel{\text{AOL}}{=} 1$$

and

$$\lim_{x \rightarrow 0^+} [x \cdot \ln(x)] = \lim_{x \rightarrow 0^+} \frac{\ln(x)}{\left(\frac{1}{x}\right)} \stackrel{\frac{-\infty}{\infty}}{=} \lim_{x \rightarrow 0^+} \frac{\left(\frac{1}{x}\right)}{\left(-\frac{1}{x^2}\right)} = \lim_{x \rightarrow 0^+} (-x) = 0$$

(c) Prove that $K = -\int_0^1 \frac{\ln x}{1+x^2} dx$.

Hint. Use HW10Q4a.

Solution:

By integration by parts where $u' = \frac{1}{1+x^2}$ and $v = \ln x$, and since $\lim_{x \rightarrow 0^+} [\arctan(x) \cdot \ln(x)] = 0$, then

$$-\int_0^1 \frac{\ln x}{1+x^2} dx = -[\arctan(x) \cdot \ln(x)]_{0^+}^1 + \int_0^1 \frac{\arctan x}{x} dx = \int_0^1 \frac{\arctan x}{x} dx \stackrel{\text{HW10Q4a}}{=} \int_0^1 \left(\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2n+1} \right) dx$$

Using term-by-term integration, we get

$$\int_0^1 \left(\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2n+1} \right) dx = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \int_0^1 x^{2n} dx = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \left[\frac{x^{2n+1}}{2n+1} \right]_0^1 = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2} = K$$

Thus $K = -\int_0^1 \frac{\ln x}{1+x^2} dx$.

5. Consider the function defined by $f(x) = \sqrt{1+x}$ for every $x > -1$.

(a) Compute the Taylor polynomial of f of order 2 about $x_0 = 0$.

Solution:

We have

$$f(x) = \sqrt{1+x} \Rightarrow f(0) = 1; f'(x) = \frac{1}{2\sqrt{1+x}} = \frac{1}{2}(1+x)^{-\frac{1}{2}} \Rightarrow f'(0) = \frac{1}{2};$$

$$f''(x) = -\frac{1}{4}(1+x)^{-\frac{3}{2}} \Rightarrow f''(0) = -\frac{1}{4}.$$

It follows that

$$T_2(x) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2 = 1 + \frac{x}{2} - \frac{x^2}{8}$$

(b) Prove that for every $x \in \mathbb{R}$ we have

$$1 + \frac{1}{2}x^4 - \frac{1}{8}x^8 \leq \sqrt{1+x^4} \leq 1 + \frac{1}{2}x^4 - \frac{1}{8}x^8 + \frac{1}{16}x^{12}$$

Solution:

We show that for every $x \geq 0$ we have

$$1 + \frac{1}{2}x - \frac{1}{8}x^2 \leq \sqrt{1+x} \leq 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3$$

and then, substituting x^4 into the above inequality yields the required result.

First, if $x = 0$ then we clearly have an equality. Else, $x > 0$ and by Taylor's remainder theorem, there exists a point $0 < c < x$ for which $R_2(x) = \frac{f'''(c)}{3!}x^3$.

Note that $f'''(x) = \frac{3}{8}(1+x)^{-\frac{5}{2}}$, and as $0 < c < x$ we have $R_2(x) \geq 0$ and

$$0 \leq \sqrt{1+x} - T_2(x) = R_2(x) = \frac{3}{8 \cdot 6(1+c)^{\frac{5}{2}}}x^3 \leq \frac{3}{48}x^3 = \frac{x^3}{16}$$

Adding $T_2(x)$ to both sides yields the result.

6. Prove or disprove each of the following statements:

(a) We have

$$\frac{4}{1 \cdot 5} + \frac{4^2}{2 \cdot 5^2} + \frac{4^3}{3 \cdot 5^3} + \frac{4^4}{4 \cdot 5^4} + \cdots = \ln(5)$$

Solution:

True. We know that $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n} = \ln(1+x)$ for every $-1 < x \leq 1$.

Using this formula for $x = -\frac{4}{5}$ we obtain

$$\frac{4}{1 \cdot 5} + \frac{4^2}{2 \cdot 5^2} + \frac{4^3}{3 \cdot 5^3} + \frac{4^4}{4 \cdot 5^4} + \cdots = \sum_{n=1}^{\infty} \frac{4^n}{n \cdot 5^n} = - \sum_{n=1}^{\infty} \frac{(-1)^{n+1} \left(-\frac{4}{5}\right)^n}{n} = -\ln\left(1 - \frac{4}{5}\right) = -\ln\left(\frac{1}{5}\right) = \ln(5)$$

(b) Consider the function defined by $f(x) = \frac{x}{1+x^4}$ for every $x \in \mathbb{R}$. Then f is 13-times differentiable at zero, and $f^{(13)}(0) = -13!$.

Solution:

True.

We note that for every $|x| < 1$ we have

$$f(x) = x \cdot \frac{1}{1+x^4} = x \cdot \frac{1}{1-(-x^4)} \stackrel{|x|<1}{=} x \sum_{n=0}^{\infty} (-x^4)^n = x \sum_{n=0}^{\infty} (-1)^n x^{4n} = \sum_{n=0}^{\infty} (-1)^n x^{4n+1}$$

It follows f is infinitely differentiable at zero, and by uniqueness of power series,

$$f^{(13)}(0) = a_{13} \cdot 13! = -13!$$

(c) Consider the function defined by $f(x) = \ln(1-x)$ for every $x < 1$, and let R_N be the Taylor remainder about $x_0 = 0$. Then there exists an $N \in \mathbb{N}$ for which $|R_N(-\frac{1}{2})| < 10^{-7}$.

Solution:

True.

We know that $\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$ for every $-1 < x \leq 1$.

Thus

$$f(x) = \ln(1-x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} (-x)^n}{n} = - \sum_{n=1}^{\infty} \frac{x^n}{n} \quad \forall -1 \leq x < 1$$

By the uniqueness theorem $f(x) = T_f(x)$ for every $-1 \leq x < 1$, thus $\lim_{N \rightarrow \infty} R_N(-\frac{1}{2}) = 0$, and therefore there exists an $N \in \mathbb{N}$ for which $|R_N(-\frac{1}{2})| < 10^{-7}$.

- (d) Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. Suppose that $\sum_{n=1}^{\infty} a_n$ is convergent.

Then $\sum_{n=1}^{\infty} \frac{a_n}{n}$ is convergent.

Solution:

True.

Consider the power series $\sum_{n=1}^{\infty} a_n x^{n-1}$. As the series $\sum_{n=1}^{\infty} a_n$ is convergent, then the power series converges for $x = 1$, and as $x_0 = 0$, then it converges at least on the interval $(-1, 1]$.

Integrating the power series term-by-term, including convergence at an endpoint where the original power series converges, then the power series $\sum_{n=1}^{\infty} a_n \frac{x^n}{n}$ is convergent for every $x \in (-1, 1]$. Thus, the series $\sum_{n=1}^{\infty} \frac{a_n}{n}$ is convergent (we substitute $x = 1$).

- (e) Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers. Suppose that $\sum_{n=0}^{\infty} a_n 3^n$ is convergent.

Then the series $\sum_{n=1}^{\infty} a_n n \cdot 3^{n-1}$ is convergent.

Solution:

False.

We choose $a_0 = 0$ and $a_n = \frac{(-1)^n}{n \cdot 3^n}$ for every $n \in \mathbb{N}$.

Thus

$$\sum_{n=0}^{\infty} a_n 3^n = \sum_{n=1}^{\infty} \frac{(-1)^n}{n \cdot 3^n} 3^n = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

is convergent (minus Leibniz).

But

$$\sum_{n=1}^{\infty} a_n n \cdot 3^{n-1} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n \cdot 3^n} n \cdot 3^{n-1} = \frac{1}{3} \sum_{n=1}^{\infty} (-1)^n$$

diverges.

- (f) Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers. Suppose that $\sum_{n=0}^{\infty} a_n 3^n$ is divergent.

Then the series $\sum_{n=1}^{\infty} a_n \cdot \frac{3^{n+1}}{n}$ is divergent.

Solution:

False.

We choose $a_n = \left(-\frac{1}{3}\right)^{n+1}$ for every $0 \leq n \in \mathbb{Z}$.

Note that

$$\sum_{n=0}^{\infty} a_n 3^n = \sum_{n=0}^{\infty} \left(-\frac{1}{3}\right)^{n+1} \cdot 3^n = \frac{1}{3} \sum_{n=0}^{\infty} (-1)^{n+1} = \text{divergent (necessary criterion)}$$

But

$$\sum_{n=1}^{\infty} a_n \cdot \frac{3^{n+1}}{n} = \sum_{n=1}^{\infty} \left(-\frac{1}{3}\right)^{n+1} \cdot \frac{3^{n+1}}{n} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \text{convergent (Leibniz)}$$

- (g) Let $f : [-1, 1] \rightarrow \mathbb{R}$ be a twice differentiable function. Suppose that $f(0) = 0$, $f'(0) = 0$ and that there exists a $K > 0$ such that $|f''(c)| \leq K$ for every $c \in [-1, 1]$. Then $\sum_{n=1}^{\infty} f\left(\frac{1}{n}\right)$ converges absolutely.

Solution:

True.

Let $T_1(x)$ be the Taylor polynomial of f of order 1 about $x_0 = 0$. Then by definition,

$$T_1(x) = f(0) + f'(0) \cdot x = 0 + 0 \cdot x = 0, \quad \forall x \in \mathbb{R}$$

It follows that

$$R_1(x) = f(x) - T_1(x) = f(x), \quad \forall -1 \leq x \leq 1$$

By Taylor's remainder theorem, for every $0 < x \leq 1$ there exists a $0 < c < x$ such that

$$R_1(x) = \frac{f''(c)}{2!} x^2$$

Since $R_1(x) = f(x)$, we obtain

$$|f(x)| = |R_1(x)| = \frac{|f''(c)|}{2!} x^2 \leq \frac{K}{2} x^2, \quad \forall 0 < x \leq 1$$

It follows that

$$\sum_{n=1}^{\infty} \left| f\left(\frac{1}{n}\right) \right| \leq \sum_{n=1}^{\infty} \frac{K}{2} \cdot \frac{1}{n^2} = \frac{K}{2} \sum_{n=1}^{\infty} \frac{1}{n^2} \stackrel{\frac{1}{2}\text{-Harmonic}}{<} \infty$$

It follows $\sum_{n=1}^{\infty} \left| f\left(\frac{1}{n}\right) \right| < \infty$ and therefore $\sum_{n=1}^{\infty} f\left(\frac{1}{n}\right)$ converges absolutely.